

An AI Attempt at a Min–Max Problem for Quadratic Forms of Signs

Exact reformulations and the bounds $1/\pi$ and $1/2$

OpenAI Codex

AI-generated research note

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Status and nonclaim. This note does not prove that the limit in Question 1 exists, and it does not prove that the limit fails to exist. It is an unsuccessful solution attempt containing several rigorous partial results. It should not be submitted or cited as a solution to the MathOverflow problem.

Abstract

For

$$F(n) = \min_{a_{ij} \in \{-1,1\}} \max_{x \in \{-1,1\}^n} \left| \sum_{1 \leq i < j \leq n} a_{ij} x_i x_j \right|,$$

we give an exact covering-radius formulation and prove

$$\frac{1}{\pi} \leq \liminf_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \limsup_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \frac{1}{2}.$$

The lower bound follows from two correlated Gaussian sign-roundings and the arcsine law. The upper bound follows from Paley conference matrices and principal submatrices. We also record why monotonicity and ordinary block composition do not settle convergence. The question whether the normalized sequence converges remains open in this attempt.

1 The question

The problem, posed on MathOverflow in 2021 [1], asks whether

$$\lim_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}}$$

exists. The emphasis is on existence, not merely on proving the $\Theta(n^{3/2})$ scale.

Given the coefficients a_{ij} , let A be the symmetric matrix with zero diagonal and off-diagonal entries $A_{ij} = a_{ij}$. Write

$$Q_A(x) = \sum_{i < j} a_{ij} x_i x_j = \frac{1}{2} x^\top A x, \quad M(A) = \max_{x \in \{-1,1\}^n} |Q_A(x)|.$$

Then $F(n) = \min_A M(A)$.

2 Switching and an exact code formulation

Changing x conjugates A by a diagonal sign matrix. Thus $M(A)$ is the largest absolute total signed edge sum in the switching class of A . The relation between switching classes and cosets of the cutset code is standard [2]; the absolute-value version needed here has the following exact form.

Proposition 1 (Augmented cut-code identity). *Put $m = \binom{n}{2}$ and*

$$c_x = (x_i x_j)_{i < j} \in \{-1, 1\}^m, \quad D_n = \{c_x : x \in \{-1, 1\}^n\} \cup \{-c_x : x \in \{-1, 1\}^n\}.$$

If $\rho(D_n)$ denotes the covering radius of D_n in the Hamming cube $\{-1, 1\}^m$, then

$$\boxed{F(n) = m - 2\rho(D_n)}.$$

Proof. For $a \in \{-1, 1\}^m$,

$$\langle a, c_x \rangle = m - 2d_H(a, c_x).$$

Because D_n contains c_x and $-c_x$,

$$\max_x |\langle a, c_x \rangle| = m - 2d_H(a, D_n).$$

Minimizing the left side over a maximizes the final distance, giving the stated formula. $\square$

In particular, the original question asks whether the second-order covering-radius deficit

$$\frac{m}{2} - \rho(D_n) = \frac{F(n)}{2}$$

has an $n^{3/2}$ -normalized limit.

3 A Gaussian-sign lower bound

The next argument is uniform over every admissible coefficient matrix.

Theorem 2. *For every $n \geq 2$,*

$$\boxed{F(n) \geq \frac{n\sqrt{n-1}}{\pi}}.$$

Consequently,

$$\liminf_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \geq \frac{1}{\pi}.$$

Proof. Fix A , let $g \sim N(0, I_n)$, and choose $t > 0$. Define

$$z^\pm = \left(I \pm \frac{t}{\sqrt{n}} A \right) g, \quad x_i^\pm = \text{sgn}(z_i^\pm).$$

Every coordinate of $z^\pm$ has variance

$$d = 1 + t^2 \frac{n-1}{n}.$$

For $i \neq j$,

$$\text{Cov}(z_i^\pm, z_j^\pm) = \pm \frac{2t}{\sqrt{n}} a_{ij} + \frac{t^2}{n} (A^2)_{ij}.$$

Let $r_{ij}^\pm$ be the corresponding correlation and set

$$u_{ij} = \frac{a_{ij}t^2(A^2)_{ij}}{nd}, \quad v = \frac{2t}{\sqrt{n}d}.$$

Then $a_{ij}r_{ij}^\pm = u_{ij} \pm v$.

For centered jointly Gaussian variables with correlation r , the arcsine law states

$$\mathbb{E}[\text{sgn}(Z_1) \text{sgn}(Z_2)] = \frac{2}{\pi} \arcsin r.$$

For completeness, write $(Z_1, Z_2) = (G_1, rG_1 + \sqrt{1-r^2}G_2)$ with independent standard Gaussians. The direction of (G_1, G_2) is uniform on the circle, and the angular measure of the two same-sign sectors gives the displayed identity. Since $\arcsin$ is odd and its derivative is at least 1 on $(-1, 1)$,

$$\begin{aligned} a_{ij} \left(\arcsin r_{ij}^+ - \arcsin r_{ij}^- \right) &= \arcsin(u_{ij} + v) - \arcsin(u_{ij} - v) \\ &\geq 2v. \end{aligned}$$

The endpoints are valid correlations, so they lie in $[-1, 1]$; the same inequality follows at an endpoint by continuity.

Summing over $m = \binom{n}{2}$ edges gives

$$\begin{aligned} \mathbb{E}Q_A(x^+) - \mathbb{E}Q_A(x^-) &= \frac{2}{\pi} \sum_{i < j} a_{ij} \left(\arcsin r_{ij}^+ - \arcsin r_{ij}^- \right) \\ &\geq \frac{4mv}{\pi} = \frac{8tm}{\pi\sqrt{n}d}. \end{aligned}$$

Both expectations belong to $[-M(A), M(A)]$. Therefore

$$M(A) \geq \frac{4tm}{\pi\sqrt{n}d}.$$

The ratio $t/(1+t^2(n-1)/n)$ is maximized at $t = \sqrt{n/(n-1)}$, where $d = 2$. Substitution yields

$$M(A) \geq \frac{2m}{\pi\sqrt{n-1}} = \frac{n\sqrt{n-1}}{\pi}.$$

The bound holds for every A , completing the proof. $\square$

Independent check. A separate finite verifier reconstructs the covariance entries for random sign matrices, checks the arcsine inequality edge by edge, checks the optimized algebra through order 9999, and includes corruption controls. This is a check of the displayed calculation, not a substitute for the proof.

4 A conference-matrix upper bound

Theorem 3.

$$\boxed{\limsup_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \frac{1}{2}.}$$

Proof. A symmetric conference matrix C of order N has zero diagonal, off-diagonal entries in $\{-1, 1\}$, and

$$C^2 = (N - 1)I.$$

For every $x \in \{-1, 1\}^N$,

$$|Q_C(x)| = \frac{1}{2}|x^\top Cx| \leq \frac{1}{2}\|x\|_2\|Cx\|_2 = \frac{1}{2}N\sqrt{N-1}.$$

The Paley construction supplies such a matrix of order $q + 1$ whenever q is a prime power with $q \equiv 1 \pmod{4}$; see, for example, [3]. The prime number theorem in arithmetic progressions allows, for each sufficiently large n , a prime $q \equiv 1 \pmod{4}$ with $q + 1 \geq n$ and $(q + 1)/n \rightarrow 1$.

Take any n -by- n principal submatrix A of the conference matrix of order $q + 1$. Compression cannot increase operator norm, so $\|A\|_{\text{op}} \leq \sqrt{q}$. Hence

$$F(n) \leq M(A) \leq \frac{1}{2}n\sqrt{q} = \left(\frac{1}{2} + o(1)\right)n^{3/2}.$$

□

5 Elementary regularity and the composition wall

Proposition 4. *For every $n \geq 2$,*

$$F(n) \leq F(n + 1) \leq F(n) + n.$$

Proof. Restrict a signing on $n + 1$ vertices to its first n vertices. For fixed $x \in \{-1, 1\}^n$, its restricted energy is the average of the two full energies obtained by setting the final sign to 1 and -1 . The restricted norm is therefore no larger than the full norm, which proves the first inequality after minimization.

For the second inequality, extend an optimal n -vertex signing by arbitrary incident signs b_i . If $L(x) = \sum_i b_i x_i$, then

$$\max_{s \in \{-1, 1\}} |Q_A(x) + sL(x)| = |Q_A(x)| + |L(x)| \leq F(n) + n.$$

□

This regularity is too weak to force convergence. A direct block construction also loses at leading order. If optimal arrays of orders n and k are joined by independently chosen cross-edge signs, a union bound gives a cross discrepancy of order

$$\sqrt{2(\log 2)nk(n + k)}.$$

When n and k are comparable, this is of order $(n + k)^{3/2}$, exactly the target scale. Thus neither Fekete's lemma nor ordinary approximate subadditivity applies.

The analogous rectangular switching game is also governed by unimodular bilinear forms and has a long discrepancy literature [4, 5]. Those bounds do not provide the lossless symmetric composition needed here.

6 What remains

Combining Theorems 2 and 3 gives the proved partial result

$$\frac{1}{\pi} \leq \liminf_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \limsup_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \frac{1}{2}.$$

Proposition 1 gives an exact second description:

$$F(n) = \binom{n}{2} - 2\rho(D_n).$$

The first missing implication is still a limit theorem. One needs either an asymptotically lossless amplification taking a near-optimal signing at one order to a dense set of larger orders, or a different interpolation that keeps the cross-block contribution at the same optimized constant. The direct block, tensor, and annealed partition-function approaches examined in this attempt all incur a leading-order loss.

Final status. This note narrows the possible asymptotic interval and supplies an exact coding formulation. It does not answer the question posed on MathOverflow.

References

- [1] A. Kontorovich, *Min-max of a quadratic form of plus-minus ones*, MathOverflow question 413935, 2021. <https://mathoverflow.net/questions/413935/>
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- [4] T. A. Brown and J. H. Spencer, *Minimization of ± 1 matrices under line shifts*, Colloq. Math. **23** (1971), 165–171.
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